

Before the Factory

Building Africa's trusted AI-enabled discovery layer for vaccine
research and development

Kozi AI

Kigali, Rwanda

May 2026

Contents

Contents	i
Words from the Co-founders	iii
I. Before the Factory	1
What is decided before manufacturing?	1
Africa's manufacturing ambition.....	1
Manufacturing capacity is not the same as supply security	2
II. The global discovery system is changing.....	3
From prediction tools to discovery systems	3
Four operating models in global discovery	4
A clinical test case: rentosertib.....	6
Why is investment moving upstream?	7
Opportunity and caution.....	8
III. Data before models.....	9
Biocuration as scientific infrastructure	9
The repository ecosystem	9
Pathogen-specific knowledge.....	9
The emerging biodata economy	10
Agent-ready biological knowledge.....	10
IV. Why the African context changes discovery.....	12
Human genomic diversity	12
Population immunology	12
Pathogen diversity and local epidemiology.....	13
Clinical, cellular, and microbiome context.....	13
Biodiversity and medicine discovery	14
V. Africa's starting assets and broken interfaces.....	15
Computational and genomic networks.....	15
Academic and research institutions	15
Translational and manufacturing assets.....	16
Regulatory and ethics networks.....	16
VI. The trusted in silico discovery layer.....	18
Defining the layer.....	18
The scientist-directed feedback loop	18
Minimum scientific requirements	18
Architecture principles	19
What the layer cannot do	20

VII. Trust, governance, and intellectual property.....21

 Scientific trust.....21

 Data governance.....21

 Intellectual property.....22

VIII. Financing the discovery layer differently.....23

 Public and philanthropic infrastructure.....23

 Milestone-based translational grants23

 Patient equity.....24

 Pull financing and procurement24

IX. Five actions.....26

X. Kozi AI and TOPE_DEEP27

References 1

Words from the Co-founders

Africa is making a consequential investment in its vaccine future. The continental ambition is for African manufacturers to develop, produce, and supply more than 60 percent of the vaccine doses required on the continent by 2040. Achieving that ambition will require more than factories. It will require a connected scientific system capable of identifying important problems, evaluating biological evidence, testing promising ideas, and advancing viable products.(1)

Many of the decisions that shape a vaccine are made before manufacturing begins. Researchers decide which pathogen to prioritize, which biological target warrants investigation, which population the product should serve, and what evidence is sufficient to justify the next experiment. The ability to participate in these decisions is therefore an important part of scientific and health security.

Artificial intelligence is beginning to change how such decisions are supported. It can help scientists interpret large bodies of biological information, compare hypotheses, identify patterns, and prioritize experiments. The World Health Organization has documented its expanding use across the development and delivery of medicines and vaccines, while also emphasizing the need for appropriate evidence, oversight, and governance.(2)

The opportunity for Africa is not simply to adopt tools developed elsewhere. It is to shape discovery systems around the pathogens, populations, institutions, and public-health priorities that matter to the continent. This will depend on representative data, curated scientific knowledge, skilled researchers, reliable computing environments, experimental partnerships, and mechanisms that preserve what is learned.

This white paper asks a practical question:

“What would it take for Africa to build a trusted, scientist-directed, AI-enabled discovery layer that turns pathogen and population knowledge into better experimental decisions and stronger vaccine research and development pipelines?”

We do not present artificial intelligence as a substitute for scientists or experimental evidence. Nor do we argue that every capability must be built by one institution or within one country. Vaccine development is inherently collaborative. The objective is to enable African institutions to contribute earlier, retain more scientific learning, and enter partnerships with stronger evidence and greater agency.

Kozi AI is developing TOPE_DEEP as one contribution to this wider effort. Our work is still evolving and will require scientific evaluation, institutional collaboration, and continued learning from researchers. This paper therefore reaches beyond our company. It is an invitation to scientists, universities, public-health institutions, laboratories, manufacturers, regulators, funders, and partners to consider what must be connected before Africa’s manufacturing ambition can become a broader discovery capability.

The factory matters. But so do the decisions made before it.

Co-founders

Kozi AI

I. Before the Factory

What is decided before manufacturing?

A vaccine does not begin with a production line. It begins with a public-health need and a series of scientific choices about the disease, the intended population, and the type of immune response a product should generate. Early research must establish a biological rationale, identify or design an appropriate antigen, select a delivery platform, and determine whether the resulting candidate warrants laboratory and preclinical investigation. (3,4)

These choices define more than the scientific concept. They influence the populations included in development, the evidence that must be generated, the form in which the vaccine could eventually be delivered, and the conditions under which it would be used. WHO product profiles formalize this connection by specifying characteristics such as indication, target population, dosing schedule, duration of protection, route of administration, safety, efficacy, presentation, storage, and shelf life. Depending on the product, they may also address platform choice, manufacturing scale, and cost of goods. (5)

The implications of these decisions extend throughout development. Antigen and platform selection affect the assays required to assess immune responses, the design of preclinical studies, the feasibility of producing consistent material, and the questions regulators will examine before human testing. The United States Food and Drug Administration, for example, requires developers seeking authorization to begin clinical trials to submit preclinical findings together with information on product quality and the technology used to manufacture the candidate. (4) Manufacturing is therefore not an isolated final step. It is progressively shaped by decisions made during discovery and early development.

Not every early choice will prove correct. Biological uncertainty remains high, and candidates may fail because the selected antigen does not induce the intended response, the platform performs differently than expected, safety concerns emerge, or the product cannot be produced reliably. The purpose of discovery is not to eliminate this uncertainty, but to generate enough credible evidence to decide which questions should proceed to more costly testing.

Control over these upstream decisions matters because they determine which health needs become research programmes, which evidence is considered relevant, and which candidates enter development. A manufacturing facility can reproduce a defined product and process. It cannot independently recover the scientific reasoning, data, intellectual property, or experimental learning through which that product was selected. The strategic value of manufacturing therefore depends partly on the discovery and development capabilities to which it is connected.

Africa's manufacturing ambition

In 2021, the African Union established the Partnerships for African Vaccine Manufacturing with the objective of enabling African manufacturers to develop, produce, and supply more than 60 percent of the vaccine doses required on the continent by 2040. The ambition was framed as the creation of a sustainable vaccine-development and manufacturing ecosystem, rather than simply an increase in production capacity. (6)

The continental Framework for Action recognises that this objective depends on several connected capabilities. Its programmes address research and development, workforce development, technology transfer and intellectual property, regulatory strengthening, investment preparation, pooled procurement, trade, and coordination across the vaccine ecosystem. (6) Africa CDC's subsequent competency frameworks further specify the scientific and technical skills required for vaccine research, development, and manufacturing. (1)

Market conditions are another part of this agenda. Vaccine facilities require sufficient and predictable demand to operate sustainably, particularly where manufacturers face high initial development and production costs. The African Vaccine Manufacturing Accelerator was launched in June 2024 as a pull-financing mechanism that can make up to US\$1.2 billion available over ten years through payments linked to eligible production and supply milestones. (7)

These measures show that Africa's manufacturing ambition is already understood as an ecosystem challenge. Production facilities must be connected to skills, technology, regulation, finance, procurement, and research capacity. The unresolved question is whether these components will mature as a sufficiently connected system to support not only the manufacture of established products, but also greater African participation in deciding which vaccine programmes are developed.

Manufacturing capacity is not the same as supply security

Africa's manufacturing pipeline is expanding, but capacity is uneven across production stages. A joint assessment by Africa CDC, the Clinton Health Access Initiative, and PATH examined 19 African manufacturers between December 2022 and March 2023. It estimated existing formulation, fill, and finish capacity at approximately 2 billion doses annually, already close to twice the continent's average annual vaccine demand. If all announced projects were completed, this capacity would exceed projected African demand in 2030 by more than twofold. In contrast, planned capacity to produce vaccine antigens remained substantially below the level required to meet continental production targets. (8)

This imbalance matters because formulation, filling, and packaging do not provide control over the active biological component of a vaccine. Manufacturers without antigen-production capability must obtain bulk drug substance, production processes, specialised materials, or technology from external originators. The same assessment found that most current and planned African manufacturing capacity did not yet have the technology-transfer agreements required to produce identified vaccine products. Excess capacity at one stage can therefore coexist with dependence and shortages at another. (8)

The commercial risk is equally significant. Vaccine facilities require reliable product portfolios, regulatory approvals, competitive production costs, and sufficient orders to keep specialised infrastructure operating. Where several manufacturers pursue the same limited markets, installed capacity may remain unused or become financially unsustainable. The 2023 assessment warned that projected excess fill-and-finish capacity could undermine the viability of individual projects unless investment, product allocation, technology transfer, and procurement are coordinated. (8)

A 2025 WHO analysis of manufacturing and procurement across 204 countries and 98 vaccine manufacturers reached a broader conclusion: regional production capacity does not automatically create supply security. Security depends on how manufacturers are connected across production stages, which technology platforms are available, whether the surrounding regulatory and market systems function, and whether countries can procure the resulting products. WHO therefore distinguishes the physical presence of manufacturing capacity from the ability of a region to obtain suitable vaccines reliably when they are needed. (9)

For Africa, the implication is precise. The success of the manufacturing agenda should not be judged only by the number of facilities announced or the maximum doses their lines could produce. It should also be judged by whether manufacturers have access to antigens and technologies, products that meet priority health needs, functioning regulatory pathways, predictable demand, and links to research programmes capable of generating future candidates. Capacity becomes supply security only when these elements operate together.

II. The global discovery system is changing

From prediction tools to discovery systems

Artificial intelligence first entered many biological workflows as a specialised predictor: a model received a defined input and produced an estimate of structure, activity, toxicity, binding, or another property. Such tools can accelerate individual tasks, but discovery requires decisions across several linked stages. A prediction becomes useful only when scientists can relate it to existing evidence, judge its uncertainty, select an experiment, interpret the result, and use that result to guide the next decision.

Protein-structure prediction shows how quickly a specialised capability can reshape research. The AlphaFold Protein Structure Database now provides open access to more than 200 million predicted protein structures, giving researchers structural information at a scale that could not have been generated experimentally. (10) This resource expands the range of proteins that can be examined for possible functions, interactions, and therapeutic relevance, including proteins from organisms important to global health.

AlphaFold 3 extended this capability beyond individual protein structures. The model predicts complexes involving proteins, nucleic acids, small molecules, ions, and modified residues within a unified framework. In published benchmarks, it improved the prediction of several interaction classes, including protein–ligand, protein–nucleic acid, and antibody–antigen complexes, relative to specialised comparison methods. (11) These advances can inform target assessment, interaction analysis, antibody research, and experimental design. They do not, by themselves, determine whether a predicted interaction occurs under relevant biological conditions or produces the desired immune or therapeutic effect.

The scale of prediction also changes the research bottleneck. When millions of structures or candidate interactions can be generated computationally, the principal challenge is no longer producing possibilities. It is deciding which predictions are sufficiently relevant and credible to test. This requires links to sequence annotation, disease biology, pathogen variation, population context, published evidence, assay design, and experimental capacity. The scientific value lies increasingly in how predictions are selected, evaluated, and converted into informative experiments.

Closed-loop research offers a more complete operating model. In a closed loop, computational analysis proposes an experiment, the experiment generates new data, and the resulting evidence updates the model or the next decision. A 2024 study combined automated synthesis, experimental characterisation, machine learning, and physics-based interpretation to investigate molecular photostability. The system explored approximately 1.5 percent of a theoretical space of 2,200 molecules while generating experimentally testable chemical insights. (12) Although this study concerned chemistry rather than vaccine development, it demonstrates the organisational shift from one-time prediction towards iterative learning between computation and experiment.

This shift does not imply that every laboratory must become fully autonomous. The more important principle is continuity. The question posed, the evidence consulted, the model and database versions used, the predictions produced, the scientist’s judgement, the experiment performed, and the resulting data should remain connected. When these elements are separated, computational outputs become difficult to reproduce, negative results are easily lost, and the same weak hypotheses may be tested repeatedly.

Global discovery is therefore evolving at two levels. Models are becoming more capable across biological and chemical tasks, while organisations are building systems that connect those models to proprietary or public data, laboratory operations, scientific review, and accumulated experimental learning. The source of advantage is shifting from prediction alone to the ability to organise a reliable cycle from evidence to hypothesis, experiment, and revision.

Four operating models in global discovery

The organisations applying AI to discovery are not all building the same type of capability. Their strategies can be grouped into four operating models: predictive engines, integrated computational and experimental platforms, data-centred discovery systems, and scientific workbenches. The distinctions matter because each model concentrates value in a different asset.

Operating model	Core asset	Typical output	Experimental connection	Main limitation
Predictive and generative engines	High-performing models	Structures, interactions, binding estimates, candidate designs	Usually depends on external laboratory programmes	Strong benchmark performance does not establish biological or clinical validity
Integrated computational and experimental platforms	Proprietary data, automated laboratories, models, and iterative workflows	Targets, compounds, experimental datasets, development programmes	Direct, repeated feedback between digital analysis and physical experiments	Expensive to build and difficult to reproduce outside the platform
Data-centred discovery systems	Differentiated biological or clinical data	Targets, patient subgroups, biological representations, model-training resources	Varies by company and partnership	Data coverage, governance, access rights, and representativeness determine usefulness
Scientific workbenches and orchestration layers	Connections among models, databases, software, compute, and researchers	Reproducible analyses, evidence packages, model runs, and workflow outputs	Can connect to external laboratories and institutional systems	Reliability depends on the quality of connected tools, data, and scientific supervision

Predictive and generative engines

The first model concentrates capability in algorithms that predict or generate biological and chemical properties. AlphaFold established the importance of this category through protein-structure prediction. Isomorphic Labs is extending the approach into a broader drug-design engine intended to predict biomolecular interactions, binding affinity, possible binding pockets, and candidate molecules. The company reported in 2026 that its latest system exceeded AlphaFold 3 on selected internal protein-ligand benchmarks, but the supporting evidence was presented as a company technical demonstration rather than as clinical validation. (13)

The strategic strength of this model is scale. A computational system can compare many more structures, interactions, or designs than scientists could investigate experimentally. Its limitation is that prediction remains separated from proof. The model may identify a plausible interaction or generate a candidate, but laboratories must still establish whether the result is biologically relevant, experimentally reproducible, safe, and useful in a product-development context.

Integrated computational and experimental platforms

The second model builds the feedback loop inside the organisation. Recursion combines automated laboratories, high-throughput cellular experiments, multimodal biological and chemical data, machine-learning models, and drug-development programmes. The company reports that its laboratories can process up to 2.2 million samples per week and that its platform contains tens of petabytes of proprietary phenomic, transcriptomic, proteomic, chemical, and clinical data. Each physical experiment is converted into digital data that can inform subsequent analysis and candidate selection. (14)

The important feature is not the volume alone. The platform is designed so that computational hypotheses lead to experiments, and experimental outcomes return to the system. This creates institutional memory and allows models, datasets, and scientific decisions to develop together. The disadvantage is the concentration of capital,

infrastructure, and knowledge required. Automated laboratories, specialised compute, proprietary datasets, and multidisciplinary teams are difficult for smaller research organisations to reproduce.

Data-centred discovery systems

A third model treats differentiated data as the principal strategic asset. Owkin applies AI to multimodal patient information, including molecular, imaging, pathological, and clinical data, to identify therapeutic targets and the patient groups most likely to benefit from them. Its TargetMATCH system, for example, is designed to produce a ranked set of targets paired with relevant patient subgroups rather than a target prediction detached from human disease heterogeneity. (15)

Basecamp Research follows a different data strategy. It is assembling genomic and environmental data from global biodiversity to broaden the biological diversity available for training foundation models. The company states that its data programmes combine genetic sequences with contextual information about the organisms and environments from which they were collected. The Trillion Gene Atlas aims to expand the scale and diversity of biological training data substantially, although its performance and downstream value will require independent scientific assessment. (16,17)

These examples show that model performance is partly determined before model training begins. The organisms, patients, environments, metadata, and experimental conditions represented in a dataset shape which biological patterns a system can detect. Data ownership and access conditions also determine who can reuse the resulting knowledge. This makes governance, consent, benefit sharing, and long-term curation part of the discovery model rather than secondary administrative concerns.

Scientific workbenches and orchestration layers

The fourth model focuses on coordinating the expanding number of scientific tools. NVIDIA BioNeMo provides models, software libraries, deployment services, and workflows for tasks including structure prediction, molecular generation, docking, property prediction, and biomolecular model development. Its architecture is modular: researchers can combine specialised models and computational services rather than relying on one system to perform every scientific task. (18)

Claude Science represents a more recent workbench approach. Anthropic describes it as an environment that connects scientific databases, coding and statistical tools, computing resources, and auditable research outputs. The purpose is to reduce fragmentation across literature search, data analysis, software environments, and scientific documentation. This remains an emerging product model, and its scientific reliability must be evaluated at the level of the connected tools and completed workflows rather than inferred from the general capabilities of the language model. (19)

Workbenches respond to a practical problem created by rapid model growth. A research team may need separate tools for literature retrieval, sequence analysis, structure prediction, docking, statistical analysis, workflow execution, and reporting. Orchestration layers attempt to connect these components while preserving inputs, outputs, model versions, and decisions. Their value lies less in owning every scientific model than in making heterogeneous tools usable within a coherent and reviewable process.

The four models are increasingly converging. Predictive-model companies are building development pipelines. Experimental platforms are developing foundation models. Data companies are adding agents and design tools. Workbench providers are connecting directly to laboratories and institutional data systems. The emerging competitive unit is therefore not a single algorithm. It is the combination of data, models, compute, experimentation, workflow design, and retained scientific learning.

A clinical test case: rentosertib

Rentosertib provides one of the clearest tests of whether an AI-enabled discovery programme can progress beyond computational prediction. Developed by Insilico Medicine for idiopathic pulmonary fibrosis, the programme used computational methods at two distinct stages: first to identify TRAF2- and NCK-interacting kinase, or TNIK, as a potential disease target, and then to design a small-molecule inhibitor against it. The resulting candidate, initially known as INS018_055, was later named rentosertib. (20)

The target-selection process combined patient-derived multiomics data with biological-network analysis, scientific literature, clinical-trial information, and other evidence to rank possible fibrosis targets. TNIK was then investigated experimentally rather than accepted on the basis of computational ranking alone. The development programme included biochemical and cellular studies, animal models of fibrosis, medicinal chemistry, pharmacology, toxicology, and two Phase 1 studies. The authors reported that the period from target discovery to nomination of a preclinical candidate was approximately 18 months. (20)

The programme subsequently entered a multicentre, double-blind, randomised Phase 2a trial conducted at 21 sites in China. Seventy-one adults with idiopathic pulmonary fibrosis were assigned to placebo or one of three rentosertib dosing groups for 12 weeks. The primary endpoint was safety and tolerability. Treatment-emergent adverse events occurred in 70.6 percent of participants receiving placebo and between 72.2 and 83.3 percent across the rentosertib groups. Treatment-related adverse events were more frequent with increasing exposure, and discontinuations included events related to liver toxicity and diarrhoea. (21)

Lung function was assessed as a secondary outcome. Mean forced vital capacity changed by -20.3 ml in the placebo group and by $+98.4$ ml in the group receiving 60 mg once daily. The study was small, lasted only 12 weeks, and was not designed to establish definitive clinical efficacy. Fifty-five of the 71 enrolled participants completed the treatment period, and several other functional outcomes showed limited or similar changes across groups. The investigators therefore concluded that the findings justified larger and longer trials, not that the medicine had been proven effective. (21)

The case is important because the candidate crossed several evidence thresholds that many AI-discovery claims do not reach. A computationally identified target was tested experimentally; a designed compound underwent preclinical evaluation; safety and pharmacokinetics were examined in humans; and the programme reached a randomised, placebo-controlled Phase 2a trial whose results were published in a peer-reviewed journal. An independent commentary described this as a clinical milestone for AI-enabled discovery while maintaining that further validation was required. (22)

The case does not establish that AI has solved drug discovery. Rentosertib has not completed Phase 3 evaluation or received regulatory approval. A single programme cannot determine whether AI-enabled candidates will achieve higher approval rates, lower total development costs, or shorter overall timelines than conventionally discovered medicines. The Phase 2a publication itself noted that AI-discovered drugs had shown failure rates in Phase 2 comparable to other candidates and that none had yet completed Phase 3 at the time of publication. (21)

It also does not provide direct evidence for vaccine discovery. Rentosertib is a small-molecule medicine for a non-communicable disease. Vaccine development must account for antigen processing and presentation, immune memory, pathogen diversity, correlates of protection, delivery platforms, and population-level use. The relevance of the case lies elsewhere: it demonstrates that an AI-originated hypothesis becomes scientifically meaningful only through a connected chain of data, target assessment, compound design, laboratory work, preclinical studies, clinical testing, and independent scrutiny.

Why is investment moving upstream?

Investment in AI-enabled discovery is increasingly directed towards assets that influence decisions before a candidate enters conventional development. These assets include proprietary biological data, foundation models, automated laboratories, scientific software, and platforms that can generate or evaluate targets and molecules. The attraction is straightforward: organisations that shape target selection, candidate design, and experimental prioritisation may capture value earlier than companies entering only after a product has already been defined.

Recent financing illustrates the scale of this expectation. Isomorphic Labs raised US\$600 million in external financing in March 2025, followed by a US\$2.1 billion Series B round in May 2026. The company stated that the later investment would support its drug-design engine, international expansion, and the progression of therapeutic programmes towards clinical development. (23) These investments demonstrate confidence in computational drug design as a strategic capability, but they remain financial commitments rather than evidence of approved products or improved clinical success rates.

Pharmaceutical partnerships provide another signal. Isomorphic Labs has established discovery collaborations with Novartis, Eli Lilly, and Johnson & Johnson across multiple targets and therapeutic modalities. The partners contribute disease biology, development experience, and product pipelines, while Isomorphic Labs contributes computational design capabilities. (24) The structure of these agreements shows that established pharmaceutical companies are seeking access to upstream platforms without assuming that computational capability can replace their own experimental, clinical, and regulatory functions.

Data-centred companies are attracting similar interest. Owkin's collaboration with Sanofi began as a €90 million strategic partnership focused on oncology target identification and patient subgrouping and was expanded in 2026 into a five-year collaboration involving specialised agents for pharmaceutical research. (25) Owkin has also built partnerships with health systems to access governed multimodal data, including pathology images, tissue resources, molecular information, and clinical metadata. These arrangements reflect a wider recognition that models trained or evaluated on differentiated patient data may support target selection, biomarker development, and clinical-trial design.

Integrated platforms are also being financed through milestone-based pharmaceutical agreements rather than equity alone. By the end of 2025, Recursion reported receiving US\$134 million from Sanofi after five programme milestones. The payments were linked to accepted discovery packages and advancing programmes in immunology and oncology. (26) This model distributes risk across stages: the platform company is rewarded when defined outputs are delivered, while the pharmaceutical partner retains development expertise and downstream decision-making.

These transactions suggest that capital is moving towards three forms of upstream control.

First, differentiated data can become a reusable discovery asset. A dataset linking biological measurements to disease context, treatment response, or environmental conditions can support multiple research programmes rather than one product.

Second, integrated platforms can generate cumulative learning. When models, experimental results, and scientific decisions remain connected, each programme may improve the infrastructure used for the next.

Third, early discovery platforms can enter several commercial relationships. They may develop internal products, license models, provide discovery services, receive milestone payments, or share future development value. This creates more than one route to returns, although each route still depends on scientific performance and contractual access to downstream development.

The investment pattern should not be interpreted as proof that AI-enabled discovery has already become more productive than established approaches. Funding reflects expectations about future value and strategic positioning. Partnerships show demand for access to a capability, not confirmation that every programme generated through it will succeed. The most meaningful evidence will come from prospectively tested candidates, reproducible experimental results, clinical progression, approvals, and products that improve health outcomes.

The strategic implication is that value in life sciences is increasingly being organised around the ability to combine data, computation, experimentation, and retained knowledge before manufacturing. Institutions that participate

only at the production stage may remain dependent on upstream decisions, even when they possess substantial physical capacity.

Opportunity and caution

The developments examined in this section point to a real change in discovery practice. Models can now analyse biological relationships, predict molecular structures and interactions, generate candidate designs, and help prioritise experiments at a scale that was previously impractical. Integrated platforms add further value by linking these capabilities to proprietary data, automated laboratories, and repeated experimental feedback. A 2025 review in *Nature Medicine* concluded that AI is influencing multiple stages of drug development, while also noting that its long-term effect on productivity and clinical success remains to be established through prospective evidence. (27)

The strongest evidence currently concerns improvements in particular tasks, not the transformation of the entire development pathway. AlphaFold has expanded access to predicted structures. Closed-loop laboratories have demonstrated more efficient exploration of selected experimental spaces. Rentosertib has progressed from AI-supported target identification and molecular design into a peer-reviewed Phase 2a trial. These are meaningful milestones, but they represent different levels of evidence and should not be combined into a single claim that AI has shortened or de-risked discovery as a whole.

Clinical progression also remains an incomplete measure. A candidate entering Phase 1 or Phase 2 has passed important preclinical and regulatory thresholds, but most of the cost, uncertainty, and evidence generation required for approval still lies ahead. Funding rounds and pharmaceutical partnerships indicate that companies expect value from upstream data and discovery platforms. They do not establish that candidates generated through these platforms will achieve higher approval rates or better health outcomes.

The transfer of these lessons to vaccines requires particular care. Machine learning can support epitope prediction, antigen selection, immune-response analysis, and the search for correlates of protection. However, vaccine performance depends on processes that are difficult to infer from sequence or structure alone, including antigen processing and presentation, immune memory, pathogen evolution, host diversity, platform effects, and the biological meaning of measured immune responses. A review of machine-learning methods for vaccine target selection therefore emphasised the value of interpretable models and the need to connect predictions to the mechanisms of vaccine-induced immunity. (28)

The opportunity is not to replace discovery with computation. It is to make parts of discovery more searchable, testable, and cumulative. The caution is that faster production of predictions can also produce more weak hypotheses, obscure uncertainty, and concentrate scientific value in organisations that control the data and experimental feedback. The decisive capability is therefore not the model alone, but the quality of the biological knowledge, evidence standards, and learning system within which it operates.

Across the four operating models, the same dependency becomes visible. Predictive engines, experimental platforms, data-centred companies, and scientific workbenches all rely on biological information that has been identified, described, linked, and maintained. The next section examines this less visible infrastructure and why discovery systems cannot become trustworthy without it.

III. Data before models

Biocuration as scientific infrastructure

Biological data do not become reliable knowledge simply because they are stored in a database. They must be linked to stable identifiers, described consistently, and traced to the experiments, publications, or computational methods from which they were derived. Biocuration provides this scientific structure.

UniProtKB shows the importance of that distinction. Expert curators assess its reviewed records, while unreviewed records are annotated mainly through automated methods. (29) The difference allows researchers to judge whether a claim about a protein is experimentally supported, curator-evaluated, or computationally inferred.

This provenance is essential for AI-enabled discovery. Biological databases may contain transferred annotations, historical interpretations, and outputs from earlier algorithms. Without tracing their origin, a model can reproduce errors or treat one prediction as independent evidence for another. (30)

AI can help curators screen literature and propose annotations, but it does not remove the need for scientific review. Current biocuration guidance identifies data quality, reproducibility, inconsistent standards, and limited domain-specific training data as continuing constraints on automation. (31)

Biocuration is therefore scientific infrastructure. It determines what a discovery system knows, how that knowledge was established, and how confidently it should inform the next experiment.

The repository ecosystem

Early vaccine discovery draws on several repositories, each answering a different scientific question. UniProtKB connects protein sequences to functional annotation and supporting evidence. (28) The Protein Data Bank archives experimentally determined molecular structures, while AlphaFold DB provides more than 200 million predicted protein structures. (10,32) These sources are complementary, but a prediction must not be presented as equivalent to an experimentally determined structure.

The Immune Epitope Database adds evidence from experimentally tested T-cell, B-cell, and MHC-ligand assays. Its records preserve details such as the antigen, host, immune response, assay, and MHC restriction, including both positive and negative findings. (33) This context matters because an epitope sequence alone does not establish relevance across populations, pathogens, or experimental conditions.

PubMed provides access to more than 40 million biomedical citations, but literature is not automatically structured knowledge. (34) Findings must still be linked to the correct pathogen, strain, protein, sequence version, assay, and evidence level before they can support candidate prioritisation.

A reliable discovery workflow must therefore connect repositories while preserving their differences. The objective is not to accumulate the largest possible volume of data, but to assemble a coherent evidence record in which every claim can be traced to its biological entity, experimental context, and source.

Pathogen-specific knowledge

General repositories describe biological entities across many organisms, but they do not fully capture the questions that matter for a particular pathogen. Discovery also requires knowledge of strain diversity, life-cycle stage, host interaction, drug resistance, immune escape, and the experimental history of individual genes or proteins.

PlasmoDB, within VEuPathDB, brings together genomic, transcriptomic, proteomic, population, and functional data for *Plasmodium* species. Researchers can examine how a gene is expressed across parasite stages, compare orthologues, inspect variation, and connect molecular evidence to malaria biology. (35) This context is important because a protein expressed in one life-cycle stage may not be relevant to an intervention intended for another.

MycoBrowser provides manually curated genomic and proteomic information for *Mycobacterium tuberculosis* and related mycobacteria, including gene function, essentiality, transcriptomic evidence, drug information, and comparative genomics. (36) Its current notice that updates have stopped because of resource constraints also illustrates a wider problem: specialist databases remain scientifically valuable but require sustained funding and stewardship to remain current.

The Los Alamos HIV databases organise annotated viral sequences, subtype and recombinant information, geographic metadata, immune epitopes, and analytical tools. (37) Such resources are essential for a rapidly evolving pathogen because candidate design must account for sequence diversity, recombination, and immune escape rather than relying on a single reference genome.

Pathogen-specific databases therefore add biological meaning that cannot be recovered from sequence alone. They help researchers ask whether a target is conserved, expressed at the relevant stage, associated with resistance, recognised by the immune system, or supported across different strains and studies. Their value lies in combining broad repositories with the specialist knowledge needed to make a defensible experimental choice.

The emerging biodata economy

Biological data are becoming strategic assets in AI-enabled discovery. Public repositories provide broad access, but many valuable datasets remain controlled because they contain sensitive human information, costly experimental measurements, proprietary annotations, or data collected under specific consent conditions.

The European Genome-phenome Archive shows how controlled access can support research without making sensitive genomic and phenotypic data openly available. Access is granted by the relevant Data Access Committee and governed by agreements specifying how approved researchers may use, store, and transfer the data. (38)

Commercial organisations are also building differentiated biological datasets. Basecamp Research is developing the Trillion Gene Atlas by combining genomic sequences with ecological and geographic context from global biodiversity partnerships. (16) Its value will depend not only on scale, but also on provenance, representativeness, access rights, and the implementation of benefit-sharing commitments.

The biodata economy therefore includes open repositories, controlled research resources, and proprietary collections. These models can coexist, but the conditions governing consent, access, reuse, and benefit sharing must remain visible. Data that cannot be traced to their source or permitted use may be technically accessible while remaining scientifically or ethically unsuitable.

Agent-ready biological knowledge

Scientific agents can search databases, run analytical tools, compare evidence, and coordinate multi-step workflows. Their reliability depends on whether the underlying knowledge is structured for both machine use and scientific review.

This requires persistent identifiers, standard vocabularies, explicit relationships, versioning, provenance, and visible access conditions. These features align with the FAIR principles, which were developed to make digital research objects findable, accessible, interoperable, and reusable. (39)

FAIR data are not automatically trustworthy. An agent must still distinguish experimental findings from predictions, identify the organism and conditions to which a claim applies, and trace each output to its source record, publication, method, and tool version. Methodological context and study limitations remain essential for responsible reuse. (40)

For discovery, the practical requirement is clear: agents should retrieve evidence through curated identifiers and preserve citations, uncertainty, intermediate steps, and human decisions. This allows scientists to inspect, correct, and reproduce the analysis.

Before the Factory

Agent-ready knowledge is curated biological evidence made traceable, interoperable, and usable within a reviewable scientific workflow.

IV. Why the African context changes discovery

Human genomic diversity

African populations contain the greatest breadth of human genetic diversity and substantial population structure shaped by migration, adaptation, admixture, and local selection. This diversity affects which variants are detected, how genetic associations are interpreted, and how accurately findings from one population can be transferred to another.

The African Genome Variation Project demonstrated this directly using dense genotype data from 1,481 individuals and whole-genome sequences from 320 individuals across sub-Saharan Africa. It identified regionally distinct patterns of genetic variation and loci associated with traits including malaria susceptibility and hypertension. The study also showed that adding diverse African haplotypes improved genotype imputation, strengthening the case for reference resources designed around African populations rather than relying mainly on datasets assembled elsewhere. (41)

Reference choice remains consequential. An evaluation involving about 11,000 participants from East, West, and Southern Africa found marked differences in imputation performance across available panels. Accuracy also varied with ancestry, including reduced concordance in genomes with greater Khoe-San ancestry. The authors concluded that geographically and ancestrally diverse African whole-genome data are still needed to improve reference panels. (42)

These limitations affect more than variant discovery. Incomplete reference data can weaken association studies, obscure rare or population-specific variants, and distort estimates of genetic contribution to disease or treatment response. A 2024 review further identified limited African representation, weak integration of genomic and clinical records, infrastructure gaps, and dependence on externally controlled systems as barriers to translating genomics into precision health. (43)

Human genomic diversity should therefore inform discovery from the beginning. It influences the biological evidence used to identify mechanisms, define populations, and judge whether a candidate developed from narrower reference groups is likely to remain relevant across African settings.

Population immunology

Genetic diversity becomes directly relevant to vaccine discovery through the human leukocyte antigen system. HLA molecules determine which pathogen-derived peptides are presented to T cells, but allele frequencies differ across populations. An epitope predicted to bind strongly to common alleles in one population may provide narrower coverage elsewhere.

A 2024 study analysed antibody responses in 2,499 infants from Uganda, Burkina Faso, and South Africa. It identified HLA associations for five of eight antigens tested across pertussis, diphtheria, hepatitis B, tetanus, *Haemophilus influenzae* type b, and measles vaccines. Fine-mapping of HLA class II regions found associations explaining up to 10 percent of antibody-response variance in the studied cohorts. (44)

The study also found differences in the genetic architecture of pertussis antibody responses between the African cohorts and an independent European-ancestry cohort. Yet computational peptide-binding analyses did not explain these differences. This is an important limitation for epitope-first discovery: predicted HLA binding can estimate one step in antigen recognition, but it does not capture HLA expression, immune regulation, prior exposure, or the full pathway leading to a measurable response. (44)

HLA is only part of the explanation. Vaccine responses also vary with age, nutritional status, co-infections, previous pathogen exposure, microbiome composition, and other environmental influences. A review of geographical differences in vaccine performance reported substantial variation across settings, including for BCG, rotavirus, yellow fever, and attenuated malaria vaccines, and concluded that these patterns arise from interacting immunological and environmental factors rather than ancestry alone. (45)

Population-aware discovery should therefore combine local HLA frequencies and binding predictions with measured immune responses and relevant exposure data. The objective is not to design separate vaccines for every population, but to identify where apparently broad candidates may have uneven recognition and where additional antigens, validation cohorts, or alternative designs are required.

Pathogen diversity and local epidemiology

A vaccine target must remain relevant across the pathogen populations and transmission settings in which the product will be used. Reference genomes and laboratory strains provide useful starting points, but they may not represent circulating lineages, recombinants, antigenic variants, or stage-specific expression in the intended population.

HIV illustrates the challenge. Its rapid mutation, recombination, and immune escape generate extensive diversity within and between infected populations. Vaccine strategies must therefore seek responses that extend across variants, often by targeting conserved regions or guiding the development of broadly neutralising antibodies. Even conserved targets require experimental confirmation because accessibility and immune recognition are not determined by sequence conservation alone. (46)

For *Plasmodium*, relevance also depends on the parasite's life cycle. Antigens expressed during the sporozoite and liver stages support vaccines intended to prevent infection, blood-stage targets aim to reduce disease, and sexual-stage antigens may interrupt transmission to mosquitoes. A biologically credible protein is therefore useful only if it is expressed at the stage that matches the intended intervention. (47)

Antigenic diversity creates a further trade-off. Highly immunogenic surface proteins may vary enough to permit strain-specific protection or immune escape, while conserved proteins may be less exposed or less capable of inducing protective immunity. Recent computational work on *P. falciparum* has therefore combined genetic diversity, stage-specific expression, essentiality, structural information, and existing immunological evidence when prioritising new antigens. (48)

Local genomic surveillance can change candidate selection by showing which variants circulate, whether proposed epitopes remain conserved, and how pathogen populations change over time. Target prioritisation should therefore be based on the pathogens observed in the intended setting, not solely on a global reference sequence.

Clinical, cellular, and microbiome context

Host and pathogen genomes do not fully describe the conditions in which immunity develops. Age, nutrition, co-infections, prior exposure, treatment history, and disease severity can alter immune states and vaccine responses. These factors often interact, which is why geographical differences in vaccine performance cannot be attributed to ancestry or HLA variation alone. (45)

Cellular context adds another layer. Bulk tissue measurements can conceal which cells express a target, which immune populations are activated, and where damaging or protective responses occur. A spatially resolved single-cell study of fatal COVID-19 in Malawi identified cellular and tissue features that differed from patterns previously reported in predominantly European and North American cohorts. The finding does not establish that every African population has a distinct response, but it shows that disease mechanisms inferred from one setting may not fully represent another. (49)

Microbiome composition may also influence immune development and vaccine responsiveness, although the evidence remains uneven. A systematic review found associations between intestinal microbial composition and humoral or cellular responses to several vaccines, but differences in study design, populations, and analytical methods limited firm causal conclusions. (50)

The 2025 African gut microbiome atlas illustrates why broader representation matters. The study analysed 1,820 adults across Burkina Faso, Ghana, Kenya, and South Africa and assembled 69,539 microbial genomes, including

many absent from existing reference catalogues. Geography was the strongest source of microbiome variation, while lifestyle and health status also contributed. (51)

These data are not yet sufficient to direct vaccine design routinely. Their immediate value is more fundamental: they show that biological reference systems built mainly from other populations can omit clinically relevant cell states, exposures, and microbial diversity. Discovery should therefore interpret genomic evidence within the clinical and biological conditions in which host and pathogen interact.

Biodiversity and medicine discovery

The continent's biological diversity includes pathogens, vectors, environmental organisms, and genetic resources that can support vaccine, diagnostic, and therapeutic research. Once biological samples or sequence data enter global databases and computational pipelines, however, scientific value can become separated from the countries, institutions, and communities from which it originated.

This creates a practical governance question. Rapid sharing of pathogen materials and genome data can accelerate surveillance and product development, but contributors also need clarity on permitted use, attribution, access to resulting knowledge, and benefit sharing. WHO guidance supports timely pathogen genome data sharing for public health while emphasising transparency, scientific recognition, and equitable participation. (52)

The distinction between physical materials and digital sequence information is increasingly important. Under the Convention on Biological Diversity, governments have established a multilateral mechanism and the Cali Fund to support benefit sharing from commercial use of digital sequence information on genetic resources. (53) For pathogens with pandemic potential, WHO Member States are separately negotiating a Pathogen Access and Benefit-Sharing system intended to connect rapid sharing of materials and sequence information with fair access to resulting vaccines, therapeutics, and diagnostics. Those negotiations were still continuing in July 2026. (54)

African biological knowledge should contribute to global discovery. Its use should also preserve provenance, recognise scientific contributors, and establish benefit-sharing conditions before commercial or public-health value is created.

V. Africa's starting assets and broken interfaces

Computational and genomic networks

Africa has already built distributed capacity to generate, analyse, and share genomic data. H3ABioNet connects 28 nodes across 17 countries and has developed computing infrastructure, analytical workflows, databases, interoperability standards, and bioinformatics training for African genomic research. (55) Its significance lies in the networked capability it created, allowing institutions to analyse complex biological data without concentrating all expertise in one centre.

Africa CDC's Africa Pathogen Genomics Initiative extends this capacity into public health. Africa PGI 2.0 supports molecular diagnostics, sequencing, epidemiology, workforce development, and data systems intended to integrate genomic surveillance into national and continental decision-making. (56) The number of African countries able to perform basic pathogen sequencing in public-health laboratories increased from seven in 2019 to 46 by 2025. (57)

The Africa Genome Archiving for Response and Insight platform, AGARI, adds a continental mechanism for secure pathogen-data sharing and analysis. Its development also includes data-curation training, recognising that sequence generation alone is insufficient without validated metadata and consistent evidence standards. (57)

Rwanda is contributing to this wider infrastructure. The Rwanda Biomedical Centre was designated an Africa CDC Regional Centre of Excellence in 2025, with the TRIBE-HUB programme supporting its development as a national and regional public-health research and training hub. (58) Rwanda is also emerging as a coordinating location for the African Genomics Centres of Excellence initiative, which aims to strengthen ethical human genomic data sharing, standardisation, bioinformatics, and translational research across participating institutions. (59)

These networks provide a strong foundation, but most were initially designed for surveillance, population genomics, training, or outbreak response. Their data do not automatically become vaccine targets or experimental programmes. The remaining task is to connect genomic observations to target conservation, antigen selection, laboratory validation, and product-development decisions.

Academic and research institutions

African universities and public research institutes already contribute capabilities required for vaccine discovery, including pathogen biology, immunology, genomics, assay development, clinical research, and postgraduate training. These strengths are distributed across institutions rather than concentrated in a single continental centre.

The Institute of Genomics and Global Health at Redeemer's University, formerly centred on ACEGID, combines infectious-disease genomics, human genomics, molecular biology, bioinformatics, and researcher training. Its work during the Ebola, Lassa fever, and COVID-19 responses showed how locally led genomic research can move from sequencing to outbreak investigation and public-health action. (60)

The Uganda Virus Research Institute provides a different part of the pathway. Its immunology programme supports immune profiling, assay validation, monoclonal-antibody research, and vaccine-trial endpoints for pathogens including HIV, Ebola, Marburg, mpox, and Rift Valley fever. UVRI also serves within CEPI's centralised laboratory network, where standardised assays are used to compare vaccine responses across development programmes. (61)

KEMRI adds extensive clinical and translational capacity. Its clinical-trials infrastructure supports Phase I to IV studies across several Kenyan sites, while its research centres have conducted vaccine studies involving HIV, tuberculosis, Ebola, yellow fever, COVID-19, and enteric pathogens. (62)

These examples show that African institutions can generate genomic evidence, investigate immune responses, validate assays, and conduct clinical studies. The limitation is continuity. Research is often organised through disease-specific grants and international collaborations, while few programmes finance the full transition from an academic finding to candidate engineering, reproducible validation, product development, and industrial partnership. Africa CDC's workforce assessment also estimates that approximately 12,500 specialised staff will be

required to meet the continent's 2040 vaccine ambition, compared with about 3,000 currently employed across vaccine research, development, and manufacturing. (1)

The missing asset is therefore not research excellence itself. It is a sustained pathway through which promising findings can acquire the technical documentation, experimental evidence, financing, and product-development ownership needed to advance beyond publication.

Translational and manufacturing assets

Africa's translational capacity is expanding beyond final-stage production. The WHO-supported mRNA Technology Transfer Programme in South Africa connects Afrigen, the South African Medical Research Council, and Biovac across technology development, research, training, and manufacturing. Afrigen developed an mRNA production platform, SAMRC supports research, and Biovac became the first manufacturer to receive the technology. By May 2025, the programme included 14 additional manufacturing partners and was extending its work towards vaccines and therapeutics for regional priority diseases. (63)

Institut Pasteur de Dakar provides another model. It combines infectious-disease research, biotechnology training, and production of a WHO-prequalified yellow fever vaccine. The MADIBA programme is intended to expand manufacturing capacity and workforce development in Senegal. Biovac is also scaling multi-vaccine manufacturing through a US\$180 million financing package announced in 2026, covering technology transfer, production expansion, and working capital. (64,65)

Rwanda is developing complementary assets. BioNTech's Kigali facility was inaugurated in June 2026 as the company's first African manufacturing site and is designed for modular production of mRNA-based products. Its long-term relevance will depend on candidate availability, regulatory approval, workforce transfer, and integration with African research priorities. (66) Rwanda also hosts the Africa Biomanufacturing Institute, selected by Africa CDC as the East African hub in a regional workforce network linking training institutions, manufacturers, research organisations, and regulators. (67) Emerging initiatives such as Bio Usawa's modular biologics facility further broaden the country's capabilities towards monoclonal-antibody development and production, although these programmes remain at different stages of validation and operation. (68)

These assets can support translation only when manufacturing expertise enters early. Candidate selection should consider process complexity, formulation, stability, quality control, cost, and scale before development choices become difficult to change. Africa's manufacturing institutions become part of discovery when they help determine not only how a product will be made, but which candidates are technically realistic to advance.

Regulatory and ethics networks

Africa already has mechanisms for coordinating the ethical and regulatory review of clinical research. The African Vaccine Regulatory Forum brings together national regulatory authorities and ethics committees to support harmonised requirements, joint reviews, scientific advice, and evaluator training across the continent. (69) AVAREF joint reviews can reduce review timelines for multinational clinical-trial applications to a maximum of 60 days, with expedited pathways targeting 25 working days and emergency reviews targeting 10 to 15 working days.

The value of AVAREF extends beyond speed. Joint reviews allow regulators and ethics committees to examine trial design, safety, candidate products, and sponsor responses together while national authorities retain responsibility for final decisions. This creates a shared scientific process and reduces duplication across countries.

The African Medicines Agency adds a continental institutional layer. Its founding treaty entered into force in November 2021, and 32 African Union Member States had ratified it by May 2026. (70) In January 2026, AMA and AUDA-NEPAD completed the transfer of continental medicines-regulatory programmes from the African Medicines Regulatory Harmonisation initiative to the Agency.

These developments do not remove differences in national regulatory capacity, legislation, staffing, or ethics-review systems. Their importance lies in creating structures through which countries can share expertise, align standards, and coordinate decisions.

Before the Factory

The remaining weakness is timing. Regulatory and ethics expertise often enters only when a clinical-trial application is nearly complete. Earlier engagement could shape evidence plans, data governance, assay validation, consent for future data use, product characterisation, and the studies required before costly development choices are fixed.

VI. The trusted in silico discovery layer

Defining the layer

A trusted in silico discovery layer is not a single model, database, or application. It is a connected capability that brings together curated biological evidence, pathogen and population context, computational methods, scientific review, experimental planning, and retained learning.

Its purpose is to support a defined research decision. For example, a team may need to determine which antigens remain conserved across circulating strains, which epitopes have relevant population coverage, or which candidates justify laboratory testing. The layer should assemble the evidence, apply appropriate tools, record uncertainty, and show how the resulting recommendation was reached.

The output is not simply a ranked list. It is an evidence record that links each candidate to its sources, assumptions, computational analyses, reviewer decisions, and experimental status. This distinction allows scientists to inspect the reasoning, challenge weak evidence, and update the conclusion when new data become available.

“In silico” therefore describes the computational part of a wider scientific process. It does not imply that vaccine discovery can be completed without laboratory, preclinical, clinical, regulatory, and manufacturing evidence.

The scientist-directed feedback loop

The process should begin with a scientific or public-health question, not with a model searching for an application.

The first step is to define the pathogen, intended population, product objective, and decision to be supported. Relevant sequence, structural, immunological, epidemiological, and experimental evidence can then be assembled and assessed for quality, provenance, and applicability.

Computational tools may be used to compare sequences, predict structures or binding, assess conservation, estimate population coverage, or rank candidates. Scientists must then review the outputs against known biology, identify unsupported assumptions, and decide which result warrants testing.

The selected experiment should include predefined success and failure criteria. Its results, including negative or inconclusive findings, should return to the evidence base and influence subsequent rankings.

Three evidence states must remain distinct:

1. A computational prediction
2. A scientist-reviewed hypothesis
3. An experimentally supported finding

WHO guidance treats AI as a means of augmenting rather than replacing human judgement and stresses human oversight, transparency, multidisciplinary review, and evidence-based evaluation. (2)

Minimum scientific requirements

Requirement	What should be recorded
Provenance	Source database, publication, sample, and evidence type
Biological identity	Organism, strain, sequence, construct, and stable identifier
Method	Model, algorithm, parameters, thresholds, and assumptions
Versioning	Database, software, model, and workflow versions
Uncertainty	Confidence, missing data, disagreement, and limitations

Scientific review	Reviewer, decision, rationale, and required follow-up
Experimental link	Assay, conditions, protocol, result, and interpretation
Reproducibility	Inputs, code, environment, outputs, and execution record
Negative evidence	Rejected candidates, failed predictions, and reasons

Provenance is necessary because a computational result cannot be evaluated without knowing where its inputs came from and how they were transformed. A review of biomedical provenance approaches found that reproducibility, data integrity, quality assessment, and protection of sensitive data were recurring requirements, while many existing implementations remained incomplete. (71)

Workflow Run RO-Crate provides one practical example of how inputs, outputs, code, workflow definitions, and execution records can be packaged together across different workflow systems. (72)

Reproducibility does not prove that a scientific conclusion is correct. It ensures that another researcher can inspect how the conclusion was produced, repeat the analysis, and identify where different assumptions lead to different results.

Architecture principles

The layer should be built around *six principles*.

Modular

Databases, models, and analytical tools should be replaceable. The system should not need to be rebuilt whenever a better method becomes available.

Model-independent

No single model is suitable for every pathogen or decision. Tools should be selected according to the scientific question and compared where necessary.

Pathogen-aware

Workflows should account for strain variation, life-cycle stage, expression, host interaction, and the intended intervention. A malaria workflow should not simply reproduce the logic of a viral vaccine workflow.

Institutionally deployable

Institutions should be able to use local, cloud, or hybrid deployment according to their computing resources, data policies, and scientific needs.

Federated where required

Sensitive genomic or clinical data should remain within the institution responsible for them where local analysis or federated methods can answer the question without transferring the underlying records. Federated learning can support multi-institutional analysis while keeping data distributed, although differences in local datasets and record structures remain important technical limitations.

Auditable and scientist-correctable

Scientists must be able to inspect sources, reject outputs, revise annotations, and record why a recommendation was changed. FAIR and machine-readable data improve interoperability, but do not remove the need to assess method quality and provenance. (39)

What the layer cannot do

The layer can help determine what should be tested and why. It cannot independently establish:

1. protective immunity;
2. biological mechanism;
3. clinical safety or efficacy;
4. manufacturability;
5. regulatory acceptability;
6. cost-effectiveness;
7. procurement demand; or
8. Market viability.

Each claim requires a different evidence pathway. Laboratory studies may support biological activity. Preclinical and clinical studies are needed to evaluate safety and efficacy. Process development and quality systems determine whether a product can be manufactured consistently. Regulators, procurers, and health systems assess whether the evidence supports approval, purchase, and use.

The role of the discovery layer is to improve the quality and traceability of decisions made before these later stages, while preserving a clear boundary between prediction, experimental support, and product evidence.

VII. Trust, governance, and intellectual property

Scientific trust

Scientific trust should attach to a claim, not to the reputation of a model, company, or institution. Researchers must be able to identify the evidence used, distinguish experimental findings from computational predictions, inspect uncertainty, and determine who reviewed the result.

Human accountability remains essential. AI may retrieve evidence, identify patterns, or rank candidates, but scientists and institutions remain responsible for interpreting outputs and deciding whether a hypothesis advances. WHO guidance identifies human autonomy, safety, transparency, accountability, inclusiveness, and sustainability as core principles for AI in health. (73) More recent WHO guidance similarly states that AI should augment rather than replace human judgement and recommends human verification and multidisciplinary oversight. (74)

Independent reproduction and experimental testing should be recorded separately from internal review. A computational result reviewed by a scientist remains a hypothesis until the relevant experiment supports it.

The governing principle is direct:

No provenance, no scientific claim. No accountable reviewer, no advancement decision.

Data governance

Data governance determines who may access biological and health data, for which purposes, under whose authority, and with what obligations.

Different data types require different controls. Public protein or pathogen databases may permit broad reuse. Human genomic and clinical data may require consent, ethics approval, controlled access, security safeguards, and restrictions on secondary use or cross-border transfer. Institutional research data, proprietary datasets, biological samples, and public-health surveillance records may each have separate legal and contractual conditions.

The African Union Data Policy Framework calls for harmonised governance, trustworthy data environments, protection of individual and collective rights, and equitable use of data for development. It does not replace national laws, ethics requirements, or institutional responsibilities. (75)

Data governance should define:

1. permitted research and commercial uses;
2. who approves access;
3. whether data may be used to train or improve models;
4. security, retention, and onward-transfer conditions;
5. attribution and publication obligations;
6. rights over derived datasets and analytical outputs;
7. benefit-sharing arrangements where relevant.

Physical localisation is not the only measure of control. Institutions may use local, controlled-access, cloud, or federated systems while retaining authority over access, reuse, model training, and resulting value.

Pathogen sharing requires additional clarity. The proposed WHO Pathogen Access and Benefit-Sharing system is intended to connect rapid access to pathogen materials and genetic sequence information with fair access to resulting vaccines, diagnostics, and therapeutics. Its annex remained under negotiation in July 2026 and should not yet be treated as settled international law. (76)

Intellectual property

Intellectual property should be addressed before research begins, particularly when universities, laboratories, data custodians, software developers, funders, and manufacturers contribute different assets.

The agreement should distinguish:

Asset	Examples
Background intellectual property	Existing software, datasets, assays, models, platforms, processes and know-how
Project results	Curated records, annotations, workflows, experimental data and candidate designs
Patentable inventions	New constructs, compositions, methods or manufacturing processes
Copyright materials	Software, documentation and original database structures
Confidential know-how	Parameters, negative results, protocols and development knowledge
Access rights	Permission to use an asset for research, validation, manufacturing or commercialisation

WIPO guidance recommends that collaborative research agreements define ownership, access rights, commercialization rights, publication, confidentiality, and the sharing of benefits and risks. These arrangements depend on contractual terms, institutional policies, funding conditions, contributions, and applicable national law. (77).

AI-enabled collaborations should state:

1. whether project data may improve a commercial or institutional model;
2. who may reuse curated annotations and workflow rules;
3. who owns candidate constructs and experimental results;
4. whether partners may continue using the workflow after the project;
5. how improvements to existing software or methods are treated;
6. who decides when to publish or file a patent;
7. what rights manufacturers receive for development and production;
8. how source institutions and communities share benefits.

No single ownership formula fits every collaboration. Joint ownership may appear equitable but can complicate licensing and enforcement. Exclusive ownership may simplify commercial development but can restrict research access. Agreements should therefore separate ownership from access and ensure that each contributor retains the rights needed to fulfil its scientific and public-interest responsibilities.

Genetic resources require additional attention. The WIPO Treaty on Intellectual Property, Genetic Resources and Associated Traditional Knowledge was adopted in 2024. Once it enters into force, it will require patent applicants in participating jurisdictions to disclose the origin or source of genetic resources, or the relevant Indigenous Peoples or local community, when an invention is based on those resources or associated traditional knowledge. The treaty had not yet entered into force by July 2026. (78)

Clear intellectual-property arrangements should enable research, validation, and product development without separating scientific value from the researchers, institutions, data custodians, and communities that helped create it.

VIII. Financing the discovery layer differently

A discovery system cannot be financed through one instrument. Shared infrastructure, experimental translation, science companies, and manufacturing face different risks and produce value on different timelines. Funding should therefore follow the stage of development and the type of asset being created.

Public and philanthropic infrastructure

Biocuration, reference datasets, secure computing, standards, shared assays, training, and interoperable workflows benefit many institutions but rarely generate immediate commercial returns. These assets should be treated as scientific infrastructure and financed primarily through governments, research funders, development institutions, and philanthropy.

This pattern already exists in neglected-disease research. WHO data covering 2007 to 2023 show that public and philanthropic sources provided most funding for vaccine research and development, reflecting the limited commercial incentive for products serving lower-income markets. (79)

Funding should cover long-term operation, not only initial construction. A database or computing platform loses value when maintenance, curation, security, technical staff, and version updates are excluded from the grant. Core infrastructure should therefore receive multi-year institutional funding, with requirements for access, governance, performance, and sustainability.

Public investment should not require every resource to be fully open. Sensitive clinical and genomic data may remain controlled, while shared standards, metadata, tools, and non-sensitive evidence can support wider use.

Milestone-based translational grants

Research grants often end after a dataset, publication, or computational prediction has been produced. Translational funding should begin where conventional academic grants often stop.

Support should be released against defined evidence milestones, such as:

1. Confirmation of target expression and conservation;
2. production of an antigen or construct;
3. validation of an assay;
4. Replication of a computational result;
5. demonstration of an immune response;
6. Assessment of manufacturability;
7. Completion of an agreed preclinical evidence package.

Funding should be stopped, redirected, or expanded according to the evidence generated. A negative result is a valid outcome when it prevents further investment in a weak candidate and is preserved for future use.

Product development partnerships provide a relevant model. They combine public or philanthropic funding with academic and industry expertise to advance vaccines, diagnostics, and therapeutics for diseases with insufficient commercial markets. WHO reports that these partnerships work across development stages and have contributed more than 60 health technologies. (80)

African translational grants should similarly fund coordinated teams rather than isolated institutions. Computational scientists, immunologists, laboratories, product developers, manufacturers, and regulators should share one development plan, evidence standard, and set of advancement criteria.

Patient equity

Grant funding is not sufficient for companies building reusable scientific platforms, proprietary workflows, candidate portfolios, or development teams. These companies require equity investors willing to accept long technical timelines, uncertain outcomes, and limited early revenue.

Patient equity should finance capabilities that can generate value across several programmes, including curated data, validated workflows, experimental partnerships, software, and institutional deployment. Investment decisions should assess the strength of the scientific team, access to laboratories and data, reproducibility, regulatory strategy, and the ability to retain learning across projects. The presence of an AI model alone is not a defensible investment case.

Public development-finance institutions can reduce risk where private investment remains limited. In 2026, Biovac secured a financing package that included a US\$20 million IFC loan and €75 million in quasi-equity from the European Investment Bank Group. Although directed towards manufacturing rather than discovery, the structure shows how long-term and risk-sharing capital can support African biotechnology assets that conventional short-term finance may not accommodate. (81)

Comparable instruments could support African discovery companies through co-investment funds, matching capital, convertible finance, and milestone-linked equity. Public participation should include conditions on scientific capacity, access, local value creation, and equitable product development rather than protecting companies from all commercial risk.

Pull financing and procurement

Manufacturers will not invest sustainably without credible demand. Procurement commitments, volume guarantees, milestone payments, and market-shaping mechanisms can reduce uncertainty after a product or production capability has reached defined standards.

The African Vaccine Manufacturing Accelerator is designed to make up to US\$1.2 billion available over ten years through incentives linked to eligible vaccine production and supply. It is a pull mechanism because payments follow verified progress towards commercially viable manufacturing rather than funding basic discovery in advance. (7)

Pooled procurement can also provide predictable volumes and reduce transaction costs. Gavi's market-shaping model combines aggregated demand and assured financing, giving manufacturers a clearer basis for production and pricing decisions.

These instruments should be informed earlier by evidence on disease burden, intended use, product characteristics, likely demand, and country willingness to adopt. WHO's Full Value of Vaccine Assessments were developed for this purpose: to connect public-health need, product development, policy, procurement, and implementation before manufacturers reach the end of an expensive development pathway. (82)

Pull financing cannot replace upstream support. Procurement incentives work only when credible products, manufacturers, regulatory pathways, and demand already exist. Discovery infrastructure, experimental translation, company formation, and manufacturing therefore require different but connected financing instruments.

Stage	Appropriate financing
Shared evidence and infrastructure	Public and philanthropic grants
Computation-to-experiment translation	Milestone-based grants
Discovery companies and reusable platforms	Patient equity and risk-sharing capital
Manufacturing and market entry	Loans, quasi-equity, procurement commitments and pull incentives

Before the Factory

The objective is not to fund every candidate through to completion. It is to ensure that strong evidence can move forward, weak programmes can stop early, shared infrastructure remains available, and viable products encounter a credible route to demand.

IX. Building the Connected Discovery Ecosystem

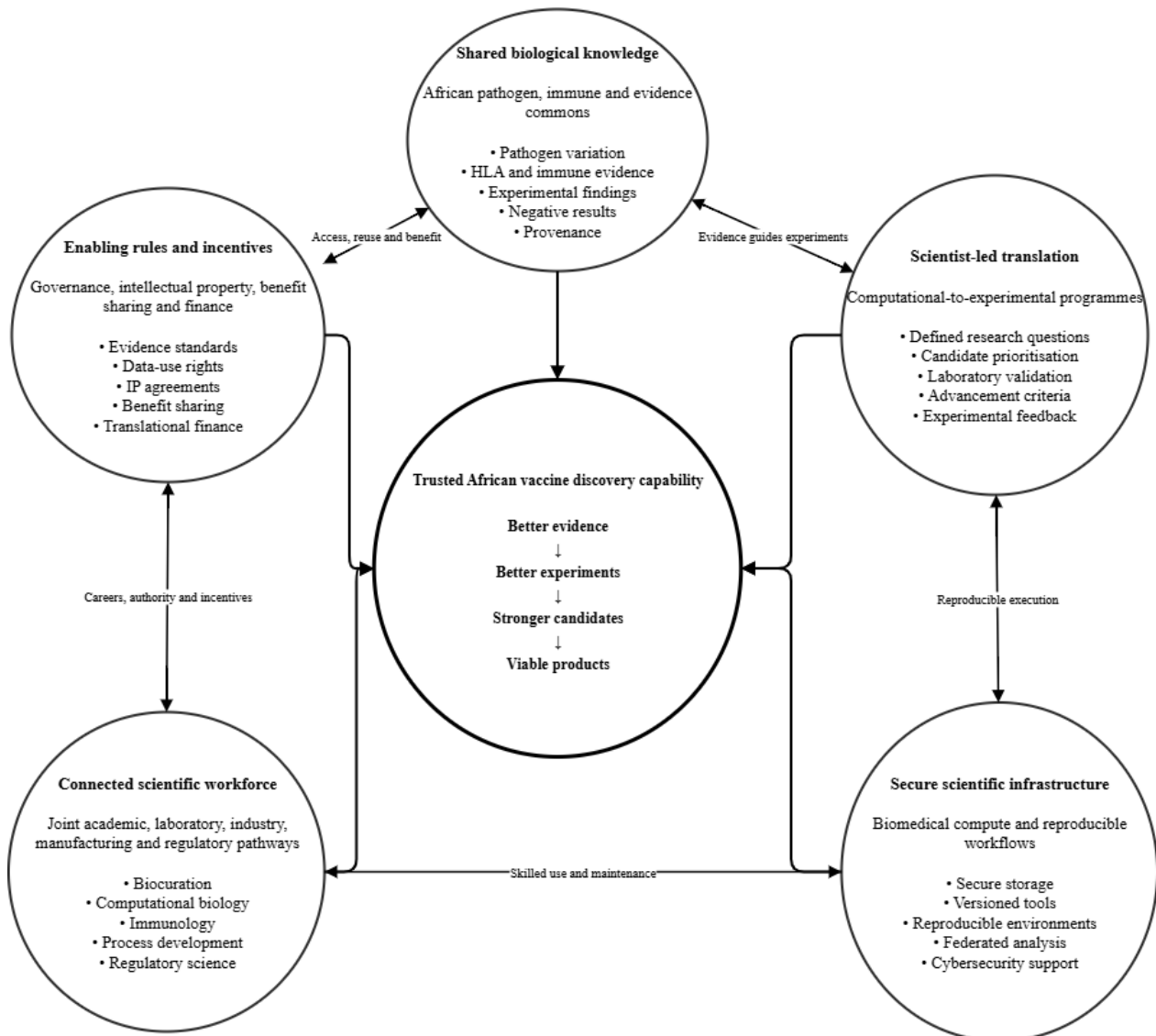


Figure 1 The connected discovery ecosystem

X. Kozi AI and TOPE_DEEP

Kozi AI is developing TOPE_DEEP as a pathogen-aware scientific evidence and orchestration platform for computational vaccinology. The platform is designed to help researchers move from a defined pathogen question to a traceable, scientist-reviewed set of hypotheses for laboratory evaluation. Its purpose is not to generate vaccines autonomously, but to connect biological evidence, analytical tools, expert judgement, and reproducible decision records within one research workflow.

The current platform integrates more than 14 biological databases and analytical tools and is structured around four stages: defining the pathogen and target population, executing coordinated computational analyses, pausing for scientist review, and producing a documented candidate report with provenance. Kozi AI reports that this approach can compress workflows that may otherwise require several weeks of manual work, although these performance claims still require independent benchmarking across pathogens, institutions, and research settings. TOPE_DEEP's initial scientific focus is epitope-first candidate prioritization. Depending on the pathogen, the workflow can examine sequence conservation, experimentally reported immune responses, predicted HLA binding and population coverage, structural evidence, toxicity, allergenicity, expression, life-cycle stage, subtype diversity, and immune escape. The platform is intended to apply pathogen-specific reasoning rather than one universal ranking model. Malaria, tuberculosis, HIV, and other pathogens require different evidence sources, biological assumptions, and validation strategies.

Kozi AI is currently working with research laboratories and biotechnology ecosystem partners to test how candidate evidence is selected, reviewed, and prepared for experimental validation. The company has publicly stated that TOPE_DEEP is being tested with laboratories working on epitope-based reverse vaccinology. These collaborations are intended to improve the scientific workflow, not to imply that the platform has already produced validated vaccine candidates. A central development priority is the experimental feedback loop. Scientists should be able to inspect the evidence behind each candidate, identify contradictory or missing findings, adjust assumptions, reject biologically implausible outputs, and record the basis of each decision. When laboratory testing occurs, the candidate record should retain the assay, material, protocol, conditions, predefined decision criteria, result, and interpretation. Positive, negative, and inconclusive findings should all update the evidence base.

Kozi AI is also mapping a federated deployment approach for institutions that cannot transfer sensitive genomic or clinical data to a central platform. Under this model, selected analyses could run within the institution holding the data, while approved outputs and provenance records are shared across the workflow. This direction is intended to support collaboration without requiring uniform data centralisation. It remains under development and will require technical testing, institutional governance agreements, and validation under different data environments. TOPE_DEEP has not established that its prioritised candidates induce protective immunity or improve vaccine-development success rates. It cannot independently determine clinical safety, efficacy, manufacturability, regulatory acceptability, procurement demand, or market viability. Its outputs depend on the quality of connected evidence, the suitability of selected tools, pathogen-specific assumptions, and scientific review. The platform's validation pathway should include reproducibility testing, retrospective comparison with known experimental findings, independent expert assessment, and prospective laboratory evaluation of candidates selected before results are known. Its credibility should be judged by whether researchers can trace its evidence, reproduce its analyses, challenge its assumptions, and test its hypotheses.

Before the Factory

Kozi AI's longer-term advantage is unlikely to come from one algorithm. It will depend on the accumulation of curated pathogen knowledge, workflow rules, scientist feedback, evaluation data, experimental partnerships, and experience deploying auditable discovery systems across institutions.

Further information is available on the [Kozi AI website](#)

References

1. Africa Centres for Disease Control and Prevention. Vaccine R&D and vaccine manufacturing competency frameworks [Internet]. Addis Ababa: Africa CDC; 2023 Nov 2 [cited 2026 Jul 17]. Available from: <https://africacdc.org/download/vaccine-rd-and-vaccine-manufacturing-competency-frameworks/>
2. World Health Organization. Benefits and risks of using artificial intelligence for pharmaceutical development and delivery. Geneva: World Health Organization; 2024. Available from: <https://www.who.int/publications/i/item/9789240088108>
3. World Health Organization. How are vaccines developed and produced? [Internet]. Geneva: World Health Organization; 2025 Feb 25 [cited 2026 Jul 17]. Available from: <https://www.who.int/news-room/feature-stories/detail/manufacturing-safety-and-quality-control>
4. US Food and Drug Administration. Vaccine development 101 [Internet]. Silver Spring (MD): US Food and Drug Administration; 2024 Aug 9 [cited 2026 Jul 17]. Available from: <https://www.fda.gov/vaccines-blood-biologics/development-approval-process-cber/vaccine-development-101>
5. World Health Organization. Preferred product characteristics and target product profiles [Internet]. Geneva: World Health Organization; [cited 2026 Jul 17]. Available from: <https://www.who.int/teams/immunization-vaccines-and-biologics/product-and-delivery-research/ppcs>
6. Africa Centres for Disease Control and Prevention. Partnerships for African Vaccine Manufacturing Framework for Action [Internet]. Addis Ababa: Africa CDC; 2022 [cited 2026 Jul 17]. Available from: <https://africacdc.org/download/partnerships-for-african-vaccine-manufacturing-pavm-framework-for-action/>
7. Gavi, the Vaccine Alliance. African Vaccine Manufacturing Accelerator [Internet]. Geneva: Gavi; [cited 2026 Jul 17]. Available from: <https://www.gavi.org/our-work/market-shaping/regional-manufacturing-strategy/avma>
8. Africa Centres for Disease Control and Prevention, Clinton Health Access Initiative, PATH. Current and planned vaccine manufacturing in Africa: results from a joint assessment by Africa CDC, CHAI, and PATH [Internet]. Addis Ababa: Africa CDC; 2023 [cited 2026 Jul 17]. Available from: <https://africacdc.org/download/current-and-planned-vaccine-manufacturing-in-africa-results-from-a-joint-assessment-by-africa-cdc-chai-and-path/>
9. World Health Organization. Global market landscape of vaccine manufacturing and procurement. Geneva: World Health Organization; 2025. Available from: <https://www.who.int/publications/i/item/9789240118249>
10. Bertoni D, Tsenkov M, Magana P, Nair S, Pidruchna I, Querino Lima Afonso M, et al. AlphaFold Protein Structure Database 2025: a redesigned interface and updated structural coverage. *Nucleic Acids Res.* 2026;54(D1):D358–D362. doi:10.1093/nar/gkaf1226.
11. Abramson J, Adler J, Dunger J, Evans R, Green T, Pritzel A, et al. Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature.* 2024;630(8016):493–500. doi:10.1038/s41586-024-07487-w.
12. Angello NH, Friday DM, Hwang C, Yi S, Cheng AH, Torres-Flores TC, et al. Closed-loop transfer enables artificial intelligence to yield chemical knowledge. *Nature.* 2024;633(8029):351–358. doi:10.1038/s41586-024-07892-1.
13. Isomorphic Labs. The Isomorphic Labs Drug Design Engine unlocks a new frontier beyond AlphaFold [Internet]. London: Isomorphic Labs; 2026 [cited 2026 Jul 17]. Available from: <https://www.isomorphiclabs.com/articles/the-isomorphic-labs-drug-design-engine-unlocks-a-new-frontier>
14. Recursion Pharmaceuticals. Technology: the Recursion OS [Internet]. Salt Lake City: Recursion Pharmaceuticals; [cited 2026 Jul 17]. Available from: <https://www.recursion.com/platform>
15. Owkin. TargetMATCH: AI-powered therapeutic target discovery [Internet]. Paris: Owkin; [cited 2026 Jul 17]. Available from: <https://www.owkin.com/targetmatch>
16. Vince O, Oldach P, Pereno V, Leung MHY, Greco C, Minto-Cowcher G, et al. Breaking through biology's data wall: expanding the known tree of life by over tenfold using a global biodiscovery pipeline [preprint]. *bioRxiv.* 2025. doi:10.1101/2025.06.11.658620.
17. Basecamp Research. Data [Internet]. London: Basecamp Research; [cited 2026 Jul 17]. Available from: <https://basecamp-research.com/bcr-data/>

18. NVIDIA. BioNeMo Framework overview [Internet]. Santa Clara (CA): NVIDIA; [cited 2026 Jul 17]. Available from: <https://docs.nvidia.com/bionemo-framework/latest/main/about/overview/>
19. Anthropic. Claude Science: an AI workbench for scientists [Internet]. San Francisco: Anthropic; 2026 [cited 2026 Jul 17]. Available from: <https://www.anthropic.com/news/claude-science-ai-workbench>
20. Ren F, Aliper A, Chen J, Zhao H, Rao S, Kuppe C, et al. A small-molecule TNIK inhibitor targets fibrosis in preclinical and clinical models. *Nat Biotechnol.* 2025;43(1):63–75. doi:10.1038/s41587-024-02143-0.
21. Xu Z, Ren F, Wang P, Cao J, Tan C, Ma D, et al. A generative AI-discovered TNIK inhibitor for idiopathic pulmonary fibrosis: a randomized phase 2a trial. *Nat Med.* 2025;31(8):2602–2610. doi:10.1038/s41591-025-03743-2.
22. Zitnik M. AI-enabled drug discovery reaches clinical milestone. *Nat Med.* 2025;31(8):2490–2491. doi:10.1038/s41591-025-03832-2.
23. Isomorphic Labs. Isomorphic Labs 2.0 [Internet]. London: Isomorphic Labs; 2026 [cited 2026 Jul 17]. Available from: <https://www.isomorphiclabs.com/press/isomorphic-labs-funding>
24. Isomorphic Labs. Isomorphic Labs enters into a research collaboration with Johnson & Johnson [Internet]. London: Isomorphic Labs; [cited 2026 Jul 17]. Available from: <https://www.isomorphiclabs.com/articles/isomorphic-labs-enters-into-a-research-collaboration-with-johnson-johnson>
25. Owkin. NHS Lothian joins Owkin's patient data network through a multi-year partnership [Internet]. Paris: Owkin; [cited 2026 Jul 17]. Available from: <https://www.owkin.com/newsfeed/nhs-lothian-joins-owkins-patient-data-network-with-multi-year-partnership>
26. Recursion Pharmaceuticals. Recursion reports fourth quarter and full-year 2025 financial results and provides business update [Internet]. Salt Lake City: Recursion Pharmaceuticals; 2026 [cited 2026 Jul 17]. Available from: <https://ir.recursion.com/news-releases/news-release-details/recursion-reports-fourth-quarter-and-full-year-2025-financial/>
27. Zhang K, Yang X, Wang Y, Yu Y, Huang N, Li G, et al. Artificial intelligence in drug development. *Nat Med.* 2025;31(1):45–59. doi:10.1038/s41591-024-03434-4.
28. Bravi B. Development and use of machine learning algorithms in vaccine target selection. *NPJ Vaccines.* 2024;9:15. doi:10.1038/s41541-023-00795-8.
29. UniProt Consortium. UniProt: the Universal Protein Knowledgebase in 2025. *Nucleic Acids Res.* 2025;53(D1):D609–D617. doi:10.1093/nar/gkae1010.
30. Pop M, Attwood TK, Blake JA, Bourne PE, Conesa A, Gaasterland T, et al. Biological databases in the age of generative artificial intelligence. *Bioinform Adv.* 2025;5(1):vbaf044. doi:10.1093/bioadv/vbaf044.
31. Arnaboldi V, Schriml LM, Jeffries M, Li P, Bello S, Sternberg P, et al. AI in biocuration: challenges, opportunities, and a roadmap for sustainable integration. *Bioinform Adv.* 2026;6(1):vbaf102. doi:10.1093/bioadv/vbaf102.
32. RCSB Protein Data Bank. PDB statistics [Internet]. [cited 2026 Jul 18]. Available from: <https://www.rcsb.org/stats/>
33. Immune Epitope Database and Analysis Resource. IEDB epitopes [Internet]. [cited 2026 Jul 18]. Available from: <https://help.iedb.org/hc/en-us/articles/114094147471-IEDB-Epitopes>
34. National Library of Medicine. About PubMed [Internet]. Bethesda (MD): National Center for Biotechnology Information; [cited 2026 Jul 18]. Available from: <https://pubmed.ncbi.nlm.nih.gov/about/>
35. Alvarez-Jarreta J, Amos B, Aurrecochea C, Bah S, Barba M, Barreto A, et al. VEuPathDB: the eukaryotic pathogen, vector and host bioinformatics resource center in 2023. *Nucleic Acids Res.* 2024;52(D1):D808–D816. doi:10.1093/nar/gkad1003.
36. Swiss Federal Institute of Technology Lausanne. MycoBrowser [Internet]. Lausanne: EPFL; [cited 2026 Jul 18]. Available from: <https://mycobrowser.epfl.ch/home/about>
37. Los Alamos National Laboratory. HIV sequence database [Internet]. Los Alamos (NM): Los Alamos National Laboratory; [cited 2026 Jul 18]. Available from: <https://www.hiv.lanl.gov/content/sequence/HIV/mainpage.html>
38. European Molecular Biology Laboratory-European Bioinformatics Institute. Accessing data in the European Genome-phenome Archive [Internet]. Hinxton: EMBL-EBI; [cited 2026 Jul 18]. Available from: <https://www.ebi.ac.uk/training/online/courses/ega-quick-tour/accessing-the-data-in-the-ega/>

39. Wilkinson MD, Dumontier M, Aalbersberg IJJ, Appleton G, Axton M, Baak A, et al. The FAIR Guiding Principles for scientific data management and stewardship. *Sci Data*. 2016;3:160018. doi:10.1038/sdata.2016.18.
40. Weissgerber TL, Gazda MA, Nilsonne G, ter Riet G, Cobey KD, Prieß-Buchheit J, et al. Understanding the provenance and quality of methods is essential for responsible reuse of FAIR data. *Nat Med*. 2024;30(5):1220–1221. doi:10.1038/s41591-024-02879-x.
41. Gurdasani D, Carstensen T, Tekola-Ayele F, Pagani L, Tachmazidou I, Hatzikotoulas K, et al. The African Genome Variation Project shapes medical genetics in Africa. *Nature*. 2015;517(7534):327–332. doi:10.1038/nature13997.
42. Sengupta D, Botha G, Meintjes A, Mbiyavanga M, Hazelhurst S, Mulder N, et al. Performance and accuracy evaluation of reference panels for genotype imputation in sub-Saharan African populations. *Cell Genom*. 2023;3(6):100332. doi:10.1016/j.xgen.2023.100332.
43. Olono A, Mitesser V, Happi A, Happi C. Building genomic capacity for precision health in Africa. *Nat Med*. 2024;30(7):1856–1864. doi:10.1038/s41591-024-03081-9.
44. Mentzer AJ, Dilthey AT, Pollard M, Gurdasani D, Karakoc E, Carstensen T, et al. High-resolution African HLA resource uncovers HLA-DRB1 expression effects underlying vaccine response. *Nat Med*. 2024;30(5):1384–1394. doi:10.1038/s41591-024-02944-5.
45. van Dorst MMAR, Pyuza JJ, Nkurungi G, Kullaya VI, Smits HH, Hogendoorn PCW, et al. Immunological factors linked to geographical variation in vaccine responses. *Nat Rev Immunol*. 2024;24(4):250–263. doi:10.1038/s41577-023-00941-2.
46. Haynes BF, Wiehe K, Borrow P, Saunders KO, Korber B, Wagh K, et al. Strategies for HIV-1 vaccines that induce broadly neutralizing antibodies. *Nat Rev Immunol*. 2023;23(3):142–158. doi:10.1038/s41577-022-00753-w.
47. Duffy PE, Gorres JP, Healy SA, Fried M. Malaria vaccines: a new era of prevention and control. *Nat Rev Microbiol*. 2024;22(12):756–772. doi:10.1038/s41579-024-01065-7.
48. Chou RT, Ouattara A, Adams M, Berry AA, Takala-Harrison S, Cummings MP. Positive-unlabeled learning identifies vaccine candidate antigens in the malaria parasite *Plasmodium falciparum*. *NPJ Syst Biol Appl*. 2024;10:44. doi:10.1038/s41540-024-00365-1.
49. Nyirenda J, Hardy OM, Silva Filho JD, Herder V, Attipa C, Ndovi C, et al. Spatially resolved single-cell atlas unveils a distinct cellular signature of fatal lung COVID-19 in a Malawian population. *Nat Med*. 2024;30(12):3765–3777. doi:10.1038/s41591-024-03354-3.
50. Ardura-Garcia C, Curtis N, Zimmermann P. Systematic review of the impact of intestinal microbiota on vaccine responses. *NPJ Vaccines*. 2024;9:254. doi:10.1038/s41541-024-01000-0.
51. Maghini DG, Oduaran OH, Olubayo LAI, Cook JA, Smyth N, Mathema T, et al. Expanding the human gut microbiome atlas of Africa. *Nature*. 2025;638(8051):718–728. doi:10.1038/s41586-024-08485-8.
52. World Health Organization. WHO guiding principles for pathogen genome data sharing. Geneva: World Health Organization; 2022. Available from: <https://www.who.int/publications/i/item/9789240061743>
53. Secretariat of the Convention on Biological Diversity. Digital sequence information on genetic resources [Internet]. Montreal: Secretariat of the Convention on Biological Diversity; 2025 [cited 2026 Jul 18]. Available from: <https://www.cbd.int/dsi-gr>
54. World Health Organization. WHO Member States agree to extend negotiations on the Pathogen Access and Benefit-Sharing annex [Internet]. Geneva: World Health Organization; 2026 May 1 [cited 2026 Jul 18]. Available from: <https://www.who.int/news/item/01-05-2026-who-member-states-agree-to-extend-negotiations-on-pathogen-access-and-benefit-sharing-annex>
55. African Genomics Data Hub. AfriGen-D Helpdesk [Internet]. [cited 2026 Jul 18]. Available from: <https://helpdesk.afrigen-d.org/helpdesk/>
56. Africa Centres for Disease Control and Prevention. Africa Pathogen Genomics Initiative: Africa PGI 2.0 [Internet]. Addis Ababa: Africa CDC; [cited 2026 Jul 18]. Available from: <https://africacdc.org/africa-pathogen-genomics-initiative-africa-pgi/>
57. Africa Centres for Disease Control and Prevention. Africa CDC launches AGARI, a continent-wide genomic data platform to strengthen outbreak response [Internet]. Addis Ababa: Africa CDC; 2025 [cited 2026 Jul 18]. Available from: <https://africacdc.org/news-item/africa-cdc-launches-agari-a-continent-wide-genomic-data-platform-to-strengthen-outbreak-response/>

58. Rwanda Biomedical Centre. RBC to strengthen its role as a Centre of Excellence following the TRIBE-HUB project launch [Internet]. Kigali: Rwanda Biomedical Centre; 2025 Aug 7 [cited 2026 Jul 18]. Available from: <https://www.rbc.gov.rw/PHSEPR/rbc-to-strengthen-its-role-as-centre-of-excellence-following-tribe-hub-project-launch/>
59. Enabel. Transforming Rwanda Biomedical Centre into an excellence hub for national and regional public health [Internet]. Brussels: Enabel; [cited 2026 Jul 19]. Available from: <https://open.enabel.be/es/RWA/2729/p/transforming-rwanda-s-biomedical-centre-into-an-excellence-hub-for-national-and-regional-public-health.html>
60. Institute of Genomics and Global Health. Institute of Genomics and Global Health Research [Internet]. Ede: Redeemer's University; 2026 Apr 17 [cited 2026 Jul 18]. Available from: <https://ighresearch.org/en/>
61. Uganda Virus Research Institute. Immunology [Internet]. Entebbe: Uganda Virus Research Institute; [cited 2026 Jul 18]. Available from: <https://uvri.go.ug/index.php/immunology>
62. Kenya Medical Research Institute. Centre for Clinical Research, Nairobi [Internet]. Nairobi: KEMRI; 2019 Sep 19 [cited 2026 Jul 18]. Available from: <https://www.kemri.go.ke/centre-for-clinical-research-ccr-nairobi/>
63. World Health Organization. mRNA Technology Transfer Programme [Internet]. Geneva: World Health Organization; [cited 2026 Jul 18]. Available from: [https://www.who.int/initiatives/mrna-technology-transfer-\(mrna-tt\)-programme](https://www.who.int/initiatives/mrna-technology-transfer-(mrna-tt)-programme)
64. Africa Centres for Disease Control and Prevention. Africa CDC, Institut Pasteur de Dakar and the South African Medical Research Council take concrete action to strengthen Africa's biomanufacturing workforce [Internet]. Addis Ababa: Africa CDC; 2023 [cited 2026 Jul 18]. Available from: <https://africacdc.org/news-item/africa-cdc-institut-pasteur-de-dakar-and-the-south-african-medical-research-council-take-concrete-action-to-ramp-up-africas-biomanufacturing-workforce/>
65. Africa Centres for Disease Control and Prevention. Statement on the Biovac-Proparco investment milestone announced during the Africa Forward Summit [Internet]. Addis Ababa: Africa CDC; 2026 [cited 2026 Jul 19]. Available from: <https://africacdc.org/news-item/statement-by-africa-cdc-on-the-biovac-proparco-investment-milestone-announced-on-the-sidelines-of-the-africa-forward-summit/>
66. Rwanda Ministry of Health. African health ministers meet in Kigali to launch BioNTech manufacturing site [Internet]. Kigali: Ministry of Health; 2026 [cited 2026 Jul 18]. Available from: <https://www.moh.gov.rw/news-detail/african-health-ministers-meet-in-kigali-to-launch-biontech-manufacturing-site>
67. Africa Centres for Disease Control and Prevention. Regional networks to strengthen Africa's vaccine and health-products manufacturing workforce [Internet]. Addis Ababa: Africa CDC; 2025 [cited 2026 Jul 18]. Available from: <https://africacdc.org/news-item/regional-networks-to-strengthen-africas-vaccine-and-health-products-manufacturing-workforce/>
68. Bio Usawa. BioMod platform deployment in Rwanda [Internet]. Kigali: Bio Usawa; [cited 2026 Jul 18]. Available from: <https://biOUSAWA.com/news-details/121/>
69. African Vaccine Regulatory Forum. Technical support and services [Internet]. Brazzaville: WHO Regional Office for Africa; [cited 2026 Jul 19]. Available from: <https://avaref.afro.who.int/support-and-services/>
70. African Medicines Agency. Ratifications [Internet]. Kigali: African Medicines Agency; [cited 2026 Jul 19]. Available from: <https://au-ama.africa/ratifications/>
71. Tiffin N, Aniebo I, Bediako Y, Djimde A, Ghansah A, Hamaluba M, et al. Six ways to empower African research and development for health. *Nat Health*. 2026;1:12–15. doi:10.1038/s44360-025-00015-8.
72. Gierend K, Krüger F, Genehr S, Hartmann F, Siegel F, Waltemath D, et al. Provenance information for biomedical data and workflows: scoping review. *J Med Internet Res*. 2024;26:e51297. doi:10.2196/51297.
73. Leo S, Crusoe MR, Rodríguez-Navas L, Sirvent R, Kanitz A, De Geest P, et al. Recording provenance of workflow runs with RO-Crate. *PLoS One*. 2024;19(9):e0309210. doi:10.1371/journal.pone.0309210.
74. World Health Organization. Ethics and governance of artificial intelligence for health: WHO guidance. Geneva: World Health Organization; 2021. Available from: <https://www.who.int/publications/i/item/9789240029200>
75. World Health Organization. Artificial intelligence and evidence-informed policy: emerging challenges and opportunities. Geneva: World Health Organization; 2026. Available from: <https://www.who.int/publications/i/item/B09667>

76. African Union Commission. AU Data Policy Framework. Addis Ababa: African Union Commission; 2022. Available from: <https://au.int/en/documents/20220728/au-data-policy-framework>
77. World Intellectual Property Organization. Technology-transfer agreements [Internet]. Geneva: WIPO; [cited 2026 Jul 20]. Available from: <https://www.wipo.int/en/web/technology-transfer/agreements>
78. World Intellectual Property Organization. WIPO Treaty on Intellectual Property, Genetic Resources and Associated Traditional Knowledge. Geneva: WIPO; 2024. Available from: <https://www.wipo.int/en/web/treaties/ip/gratk/index>
79. World Health Organization. Research and development funding flows for neglected diseases by disease, year and funding category [Internet]. Geneva: World Health Organization; 2025 [cited 2026 Jul 20]. Available from: <https://www.who.int/observatories/global-observatory-on-health-research-and-development/monitoring/r-d-funding-flows-for-neglected-diseases-by-disease-year-and-funding-category>
80. World Health Organization. Product development partnerships [Internet]. Geneva: World Health Organization; [cited 2026 Jul 20]. Available from: <https://www.who.int/our-work/science-division/research-for-health/product-development-partnerships>
81. International Finance Corporation. IFC, EIB Group and European Commission back Biovac to build Africa's first end-to-end multi-vaccine manufacturing site, advancing vaccine self-reliance [Internet]. Washington (DC): IFC; 2026 [cited 2026 Jul 20]. Available from: <https://www.ifc.org/en/pressroom/2026/ifc-eib-group-and-european-commission-back-biovac>
82. World Health Organization. Full Value of Vaccine Assessments [Internet]. Geneva: World Health Organization; [cited 2026 Jul 20]. Available from: <https://www.who.int/teams/immunization-vaccines-and-biologicals/product-and-delivery-research/full-vaccine-value-assessments>